In fact, the enzyme-substrate union depends on a reciprocal fit between the structure of the enzyme and the substrate. Since two molecules (the substrate and the enzyme) are involved, this hypothesis is also known as the concept of intermolecular fit. The enzyme-substrate complex is highly unstable and almost immediately this complex decomposes to form the product. (Fig 9.7).

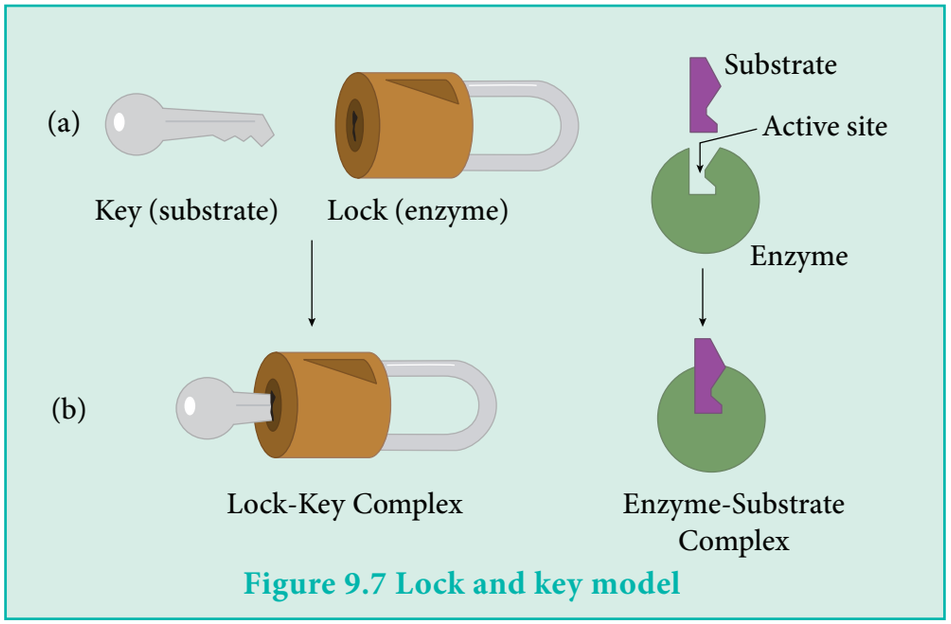

Figure 9.7 Lock and key model

(ii) Koshland's induced fit model: According to this model, the initial interaction between enzyme and substrate is relatively weak. These weak interactions quickly induce conformational changes in the enzyme that strengthen binding. This is the more accepted model for enzyme-substrate complex than the lock-and-key model. The failure of Fischer's model is the rigidity of the active site. Unlike the lock-and-key model, the induced fit model shows that enzymes are flexible structures in which the active site continually changes by interacting with the substrate (Fig 9.8).

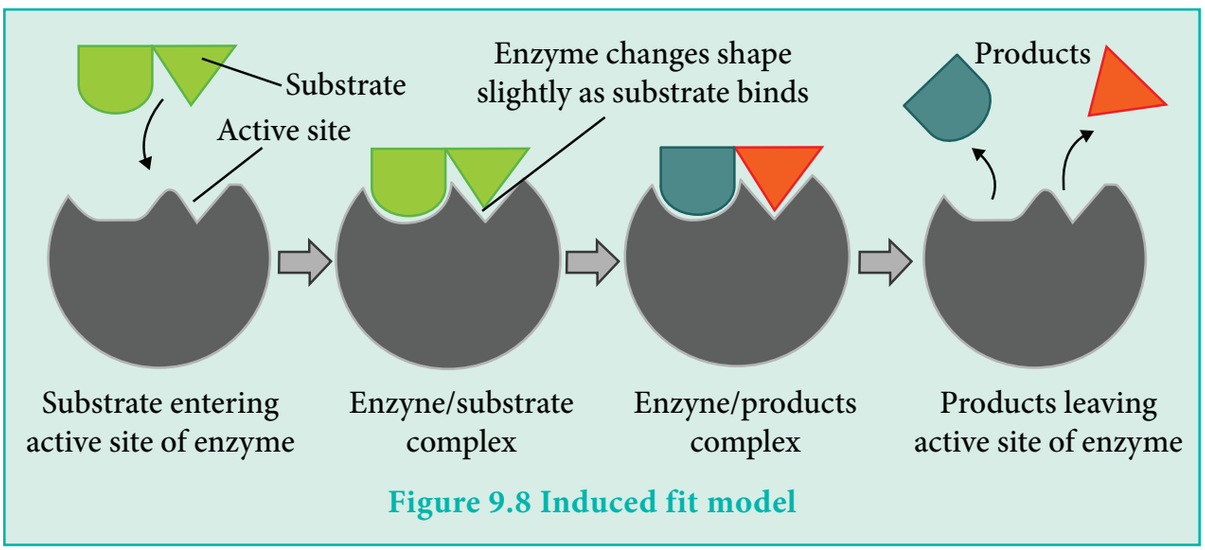

Figure 9.8 Induced fit model

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